PyGameMaker — Tutorial 5: Sokoban

How to Use This Handout

This handout goes with the in-app tutorial **Help > Tutorials > Sokoban: Box-Pushing Puzzle** (4 pages, about 20-25 minutes). Keep the tutorial open on one half of your screen. Tick each box when the step is done, and check the **You should see** line at the end of every phase. If something does not match, look at **Stuck?** before you call your teacher.

What You Will Make

A classic puzzle: you walk on a grid, one square at a time, and push crates onto target spots. A crate turns green when it is on a target, and you can press **R** to restart the level.

Phase 1: Player and Walls

- ☐ **1.** Create two sprites (32×32): spr_player and spr_wall.
- ☐ **2.** Create obj_wall (sprite spr_wall, **Solid**) and obj_player (sprite spr_player).
- ☐ **3.** In obj_player add four **Key press** events (Right, Left, Up, Down arrows). Each does *Move grid* in its direction with size **32**.
- ☐ **4.** Add **Collision with obj_wall** with *Stop movement*.
- ☐ **5.** Create room_sokoban (320×320, a 10×10 grid), turn on grid snap 32×32, put walls around the edge and a few inside, and place the player.

> **You should see:** press **F5**. Each key press moves you exactly one square. Walls stop you.

Phase 2: Pushing Crates

- ☐ **6.** Create a sprite spr_crate and an object obj_crate (sprite spr_crate, **Solid**).
- ☐ **7.** In obj_player add **Collision with obj_crate**: *If can push (facing)* then *Push other instance 32*.
- ☐ **8.** In obj_crate add **Collision with obj_wall** with *Stop movement*.
- ☐ **9.** Place 2 or 3 crates in the room, each with room to move (not in a corner!).

> **You should see:** press **F5**. Walk into a crate and it moves one square. It does not move if a wall or another crate is behind it.

Phase 3: Targets and Controller

- ☐ **10.** Create sprites spr_target and spr_crate_ok (a green crate).
- ☐ **11.** Create obj_target (sprite spr_target) — **not Solid**, so crates and the player can go over it.
- ☐ **12.** In obj_crate add a **Step** event: *If colliding with obj_target*, *Set sprite to spr_crate_ok*; *Else*, *Set sprite to spr_crate*.
- ☐ **13.** Create obj_controller (no sprite). **Draw** event: *Draw text* "Push crates onto targets!" at x 10, y 10. **Key press R**: *Restart room*.
- ☐ **14.** In the room, place the targets **first**, then **delete the player and the crates and place them again**. Place obj_controller anywhere. The number of targets must equal the number of crates.

You should see: press **F5**. Push a crate onto a target and it turns green; push it off and it turns brown again. **R** puts everything back.

Why place the targets first? The game draws things in the order you placed them. A target placed after a crate or the player is drawn on top of them, so a crate on a target would stay hidden under the red mark.

Stuck?

What is wrong	Most likely cause	What to do
I move several squares at once, or not at all	The events are *Keyboard* (held) instead of *Key press*, or *Move grid* size is not 32	Use **Key press** events and *Move grid* size 32
I walk through walls	obj_wall is not Solid, or the player has no **Collision with obj_wall**	Tick **Solid**; add the event with *Stop movement*
The crate does not move	The player has no collision event for the crate, or the crate is not Solid	Add **Collision with obj_crate** with *If can push*; tick **Solid** on the crate
The crate moves through walls	The crate has no **Collision with obj_wall** event	Add it with *Stop movement*
The crate never turns green	The Step event is missing, or the target is drawn over the crate	Add the Step event; place the targets first, then place the crates again
I cannot push a crate onto a target	obj_target is Solid	Untick **Solid** on obj_target
The crate is stuck in a corner at the start	It was placed in a corner	Place crates with room to move
The text is hard to read	It is black text over the dark room	Look at the top row, or ask your teacher about *Set draw color*

Challenges

- **Try this (5 minutes):** draw a nicer player or crate.
- **Push further:** design a harder level with tighter corridors; check you can solve it yourself first.
- **Invent:** count the moves with a variable, or make several rooms and go to the next one when all crates are on targets.

Vocabulary

Term	What it means
Grid movement	Moving in fixed steps (here, 32 pixels) instead of smoothly
Key press	An event that happens once when a key goes down
Solid	An object that blocks movement
Target	A marker on the floor; not solid

Restart room	Put the level back the way it started

Check Yourself

Write your answers in the notes below.

1. Why do we use **Key press** events and not *Keyboard (held)* for a Sokoban player?
2. Why must the targets not be Solid?
3. Why is a crate not pushed if there is another crate behind it?

My Notes